DIGITAL IMAGE PROCESSING

LECTURE # 11

COLOR IMAGE PROCESSING-II

29th April, 2026

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Topics to Cover

- Pseudo Color Processing
- Intensity Slicing
- Gray to Color Conversion
- Full Color Image Processing
- Color Complements
- Color Slicing
- Color Correction
- Tone Correction
- Histogram Processing
- Color Image Smoothing
- Color Image Sharpening
- Noise in Color Images
- Color Segmentation
- Color Image Compression

False Color

- A **false-color image** is an image that depicts an object in colors that differ from those a photograph (a **"true-color" image**) would show.

- To understand **false color**, a look at the concept behind **true color** is helpful.

- An image is called a *"true-color" image* when it offers a natural color rendition, or when it comes close to it.

- This means that the colors of an object in an image appear to a human observer the same way *as if* this observer were to *directly* view the object: A green tree appears green in the image, a red apple red, a blue sky blue, and so on.

- When applied to black-and-white images, *true-color* means that the perceived lightness of a subject is preserved in its depiction.

- A **false-color image** sacrifices natural color rendition in order to ease the detection of features that are not readily visible otherwise – for example, the use of near infrared for the detection of vegetation in satellite images.

Pseudo-Color Image Processing

- Pseudo-color Image Processing consists of assigning colors to gray levels based on specific criterion

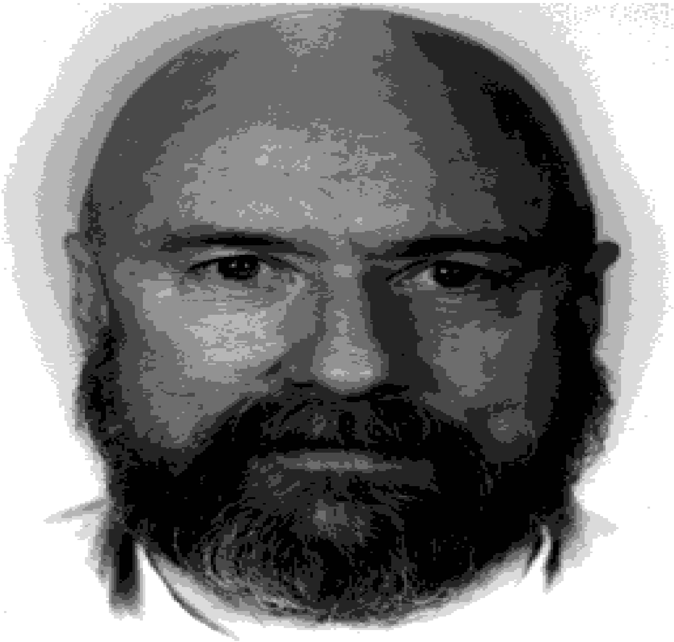
- Generally, the eye cannot distinguish more than about 2 dozen gray levels in an image. Thus, subtle detail can easily be lost in looking at gray scale images.

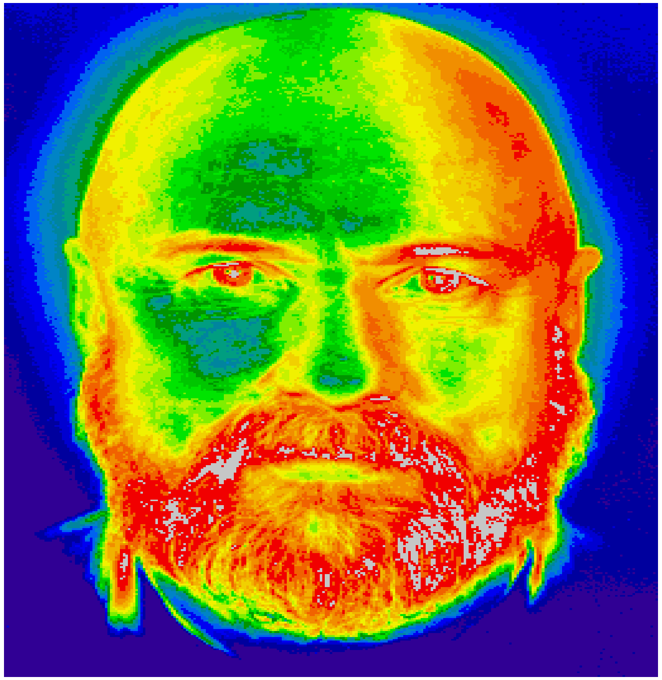
- To enhance variations in gray level and make them more obvious, gray scale images are frequently pseudo-colored, where each gray scale (generally at least 256 levels for most displays) are mapped to a color level through a look-up table (LUT).

- The eye is extremely sensitive to color and can distinguish thousands of color values in a picture.

Pseudo-Coloring using LUT

- CLUT(Color lookup table):: A mapping of a pixel value to a color value shown on a display device.
  - For example, in a grayscale image with levels 0, 1, 2, 3, and 4, pseudo-coloring is a color lookup table that maps 0 to black, 1 to red, 2 to green, 3 to blue, and 4 to white.





Intensity Slicing

- The technique of intensity slicing or density slicing, or color coding is one of the simplest example of Pseudo-color image processing

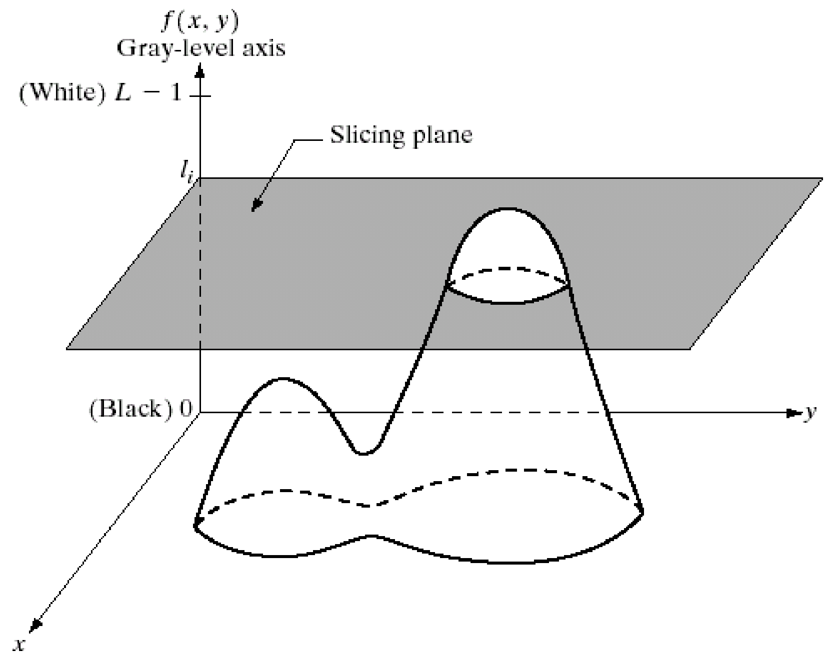


FIGURE 6.18 Geometric interpretation of the intensity-slicing technique.

Intensity Slicing

- The Gray Scale [0,L-1] is divided into L levels; where l_0 represents Black $(f(x,y)=0)$ and l_{L-1} represents white $(f(x,y)=L-1)$

- Suppose that P planes perpendicular to the intensity axis are defined at levels $l_1, l_2, \ldots, l_p$

- Then assuming that $0<P<L-1$ the P planes partition the gray scale into P+1 intervals, $V_1, V_2, \ldots, V_{p+1}$

Intensity Slicing

- Gray level to color assignments are made according to the relation:

$$f(x,y)= c_k \quad \text{if } f(x,y) \in v_k$$

- Where c_k is the color associated with the kth intensity interval v_k defined by the partition planes at l=k-1 and l=k

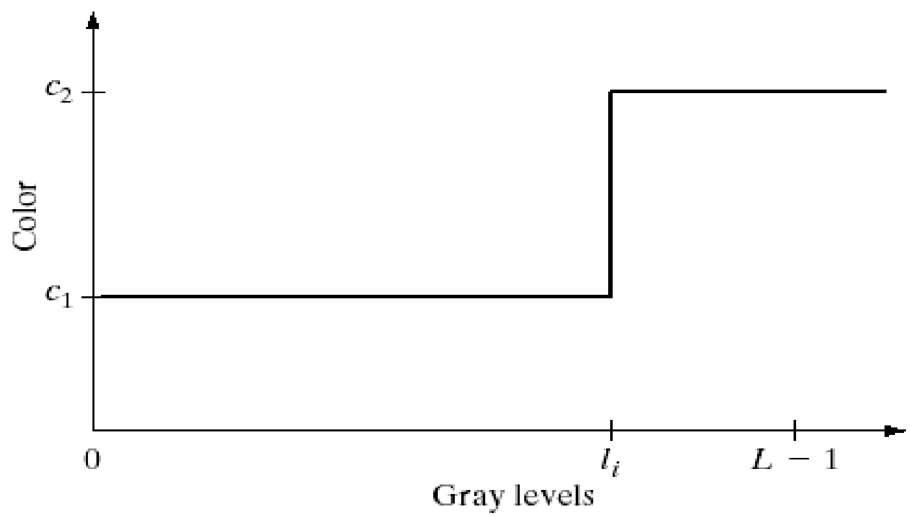
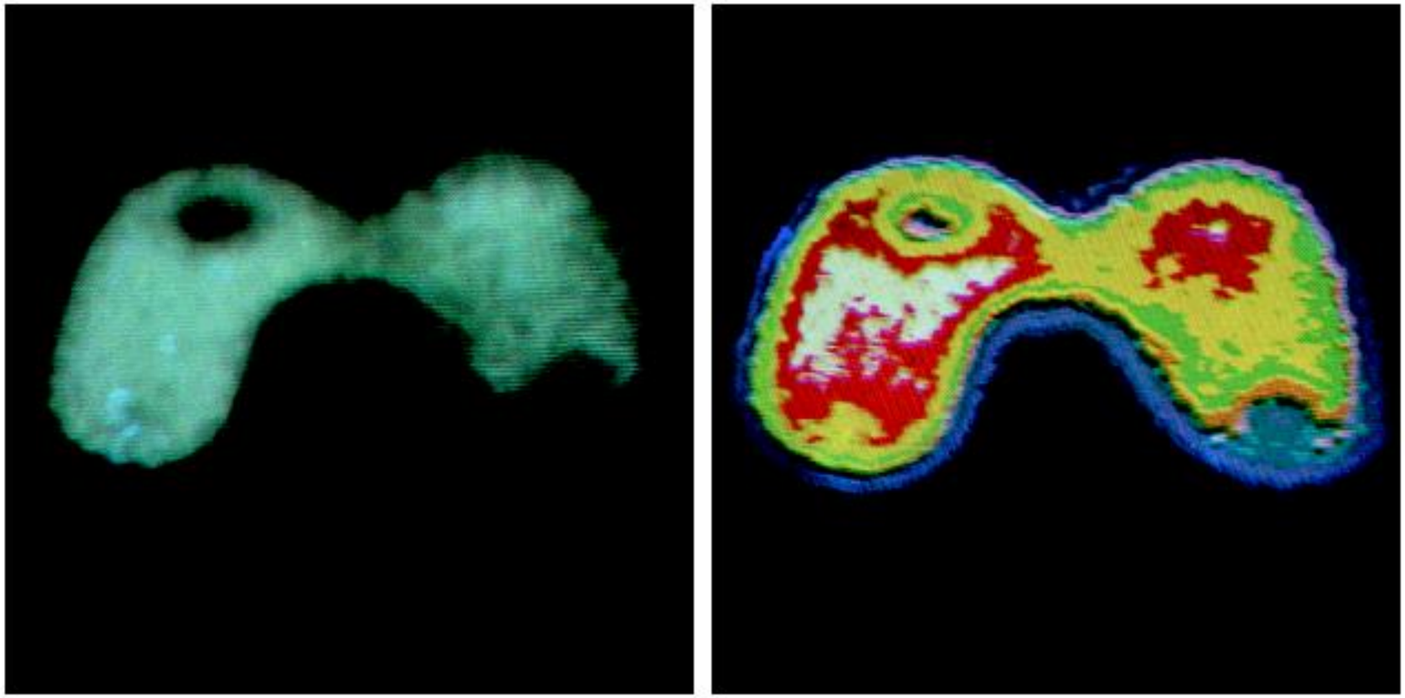


FIGURE 6.19 An alternative representation of the intensity-slicing technique.

Intensity Slicing



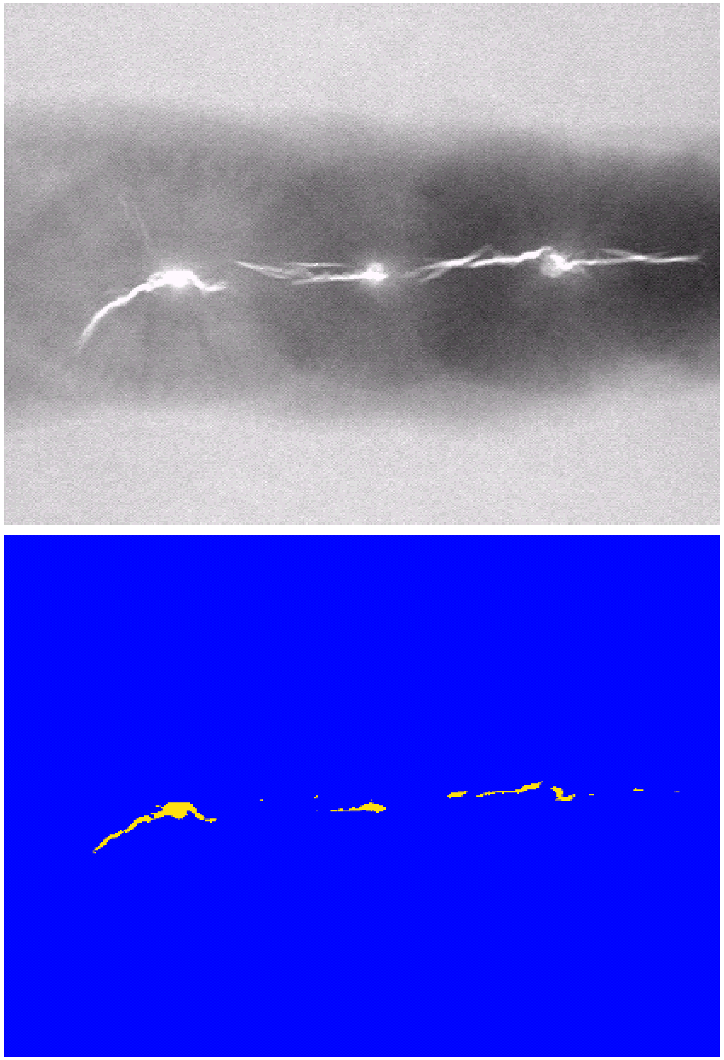
a b

FIGURE 6.20 (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

Intensity Slicing

a
b

FIGURE 6.21
(a) Monochrome X-ray image of a weld. (b) Result of color coding. (Original image courtesy of X-TEK Systems, Ltd.)

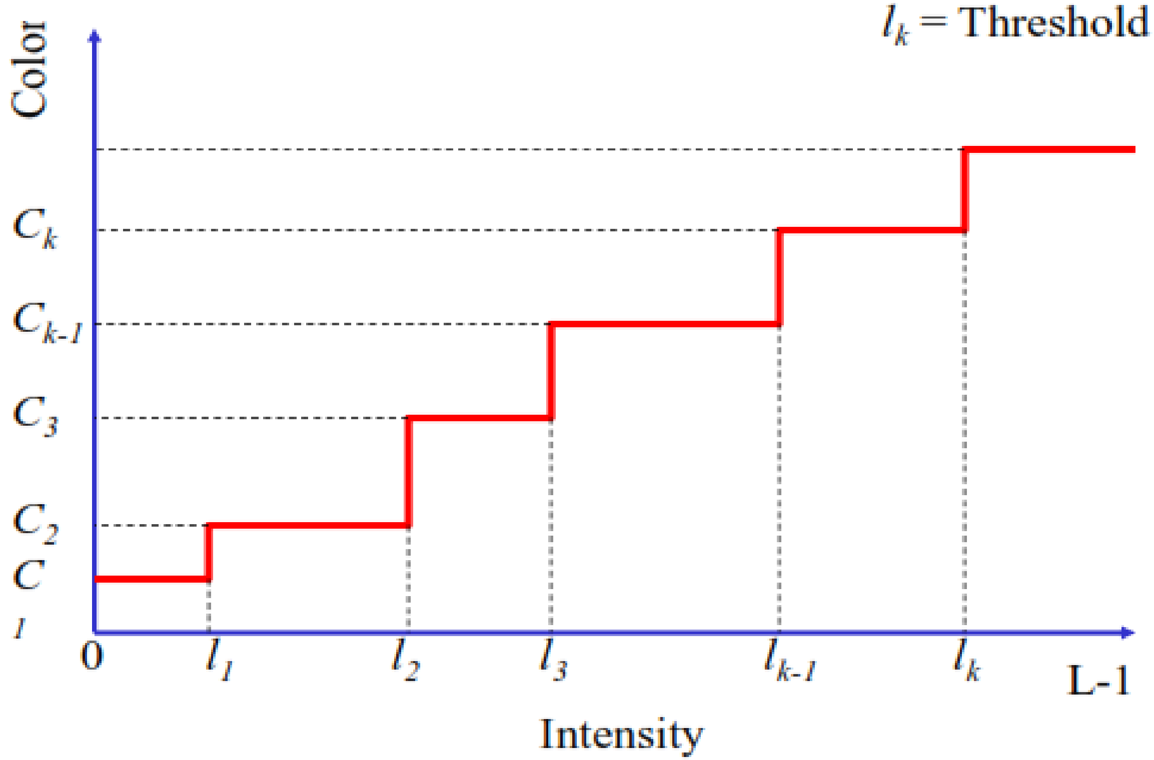


Multi Level Intensity Slicing

$$g(x,y) = C_k \qquad \text{for } l_{k-1} < f(x,y) \le l_k$$

grey level:	0–63	64–127	128–191	192–255
colour:	blue	magenta	green	red

C_k = Color No. k
l_k = Threshold level k



Gray to Color Conversion

- In this method, a given gray level image is processed by 3 different transformation functions producing 3 enhanced images in Red, Green and Blue channels respectively.

- By combining the 3 channel images, we get a colored image.

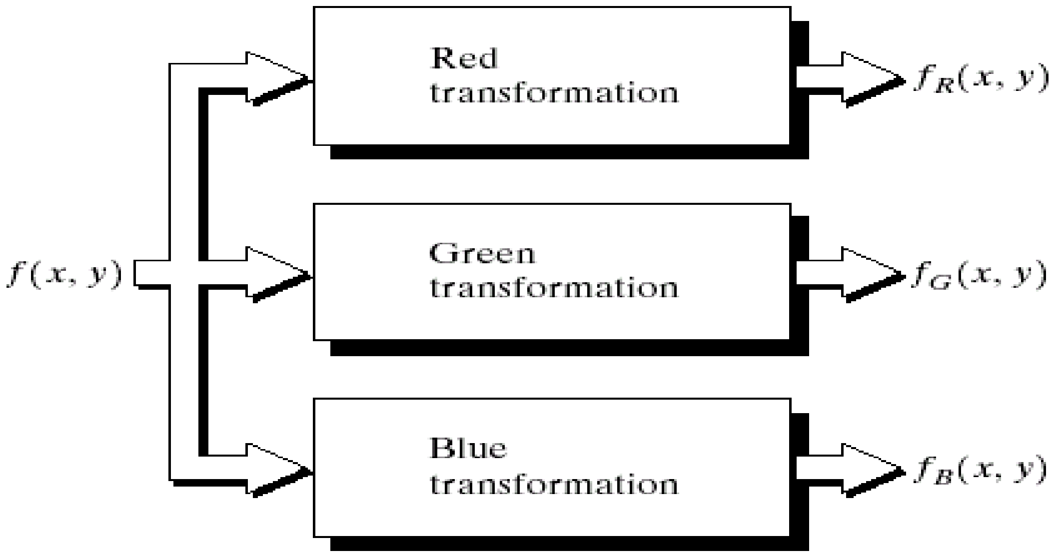


FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

Gray to Color Conversion

$$f_R(x, y) = f(x, y)$$

$$f_G(x, y) = 0.33\, f(x, y)$$

$$f_B(x, y) = 0.11\, f(x, y)$$

Gray to Color Conversion

$$f_R(x, y) = 0.33\, f(x, y)$$

$$f_G(x, y) = f(x, y)$$

$$f_B(x, y) = 0.11\, f(x, y)$$

Gray to Color Conversion

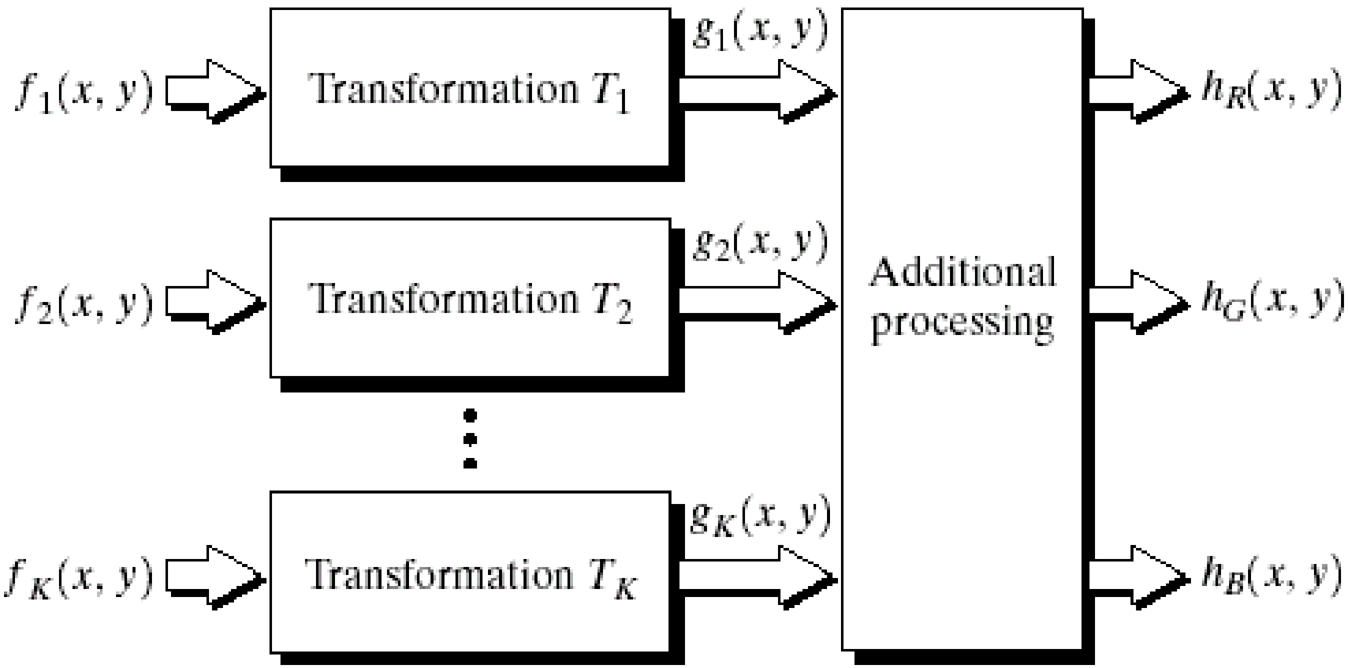
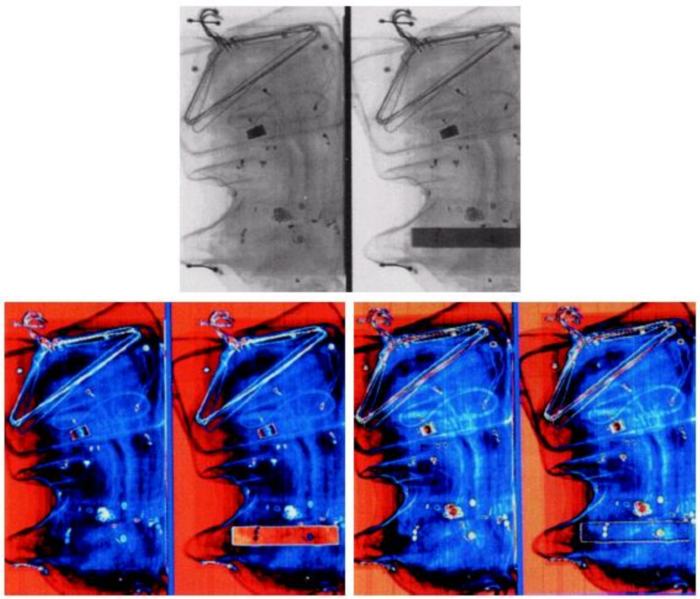


FIGURE 6.26 A pseudocolor coding approach used when several monochrome images are available.

Gray to Color Conversion

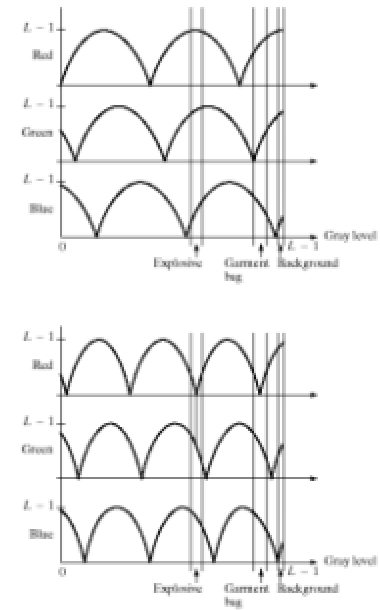


a
b c

FIGURE 6.24 Pseudocolor enhancement by using the gray-level to color transformations in Fig. 6.25. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)

produced image (b) in the previous example.

produced image (c) in the previous example.



a
b

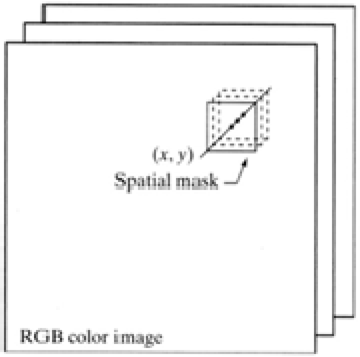
FIGURE 6.25 Transformation functions used to obtain the images in Fig. 6.24.

note the phase and frequency changes in the transformations produces different colors

Basics of Full Color Image Processing

- Full color image processing fall into 2 categories.
  - In 1st category we process each component image individually and then form a composite processed color image from the individually processed component.
  - In 2nd category we work with color pixels directly. Because full color images have at least three components, color pixels are really vectors.
  - Let c represent an arbitrary vector in RGB color space:

$$C = \begin{bmatrix} C_r \\ C_g \\ C_b \end{bmatrix} = \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



Basics of Full Color Image Processing

- Color components are the function of co-ordinates(x,y) so we can write it as:

$$C(x,y) = \begin{pmatrix} C_R(x,y) \\ C_G(x,y) \\ C_B(x,y) \end{pmatrix} = \begin{pmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{pmatrix}$$

- For an image of size MxN there are MN such vectors, c(x,y), for x=0,1,2,…,M-1; y=0,1,2,…,N-1

Color Transformations

- Color transformation can be represented by the expression ::

  $g(x,y)= T\,[f(x,y)]$
  $f(x,y)$: input image
  $g(x,y)$: processed (output) image
  $T[*]$: an operator on f defined over neighborhood of (x,y).

- The pixel values here are triplets or quartets (i.e group of 3 or 4 values)

Color Transformations

- $S_i = T_i(r_1, r_2, \ldots, r_n) \qquad i = 1, 2, 3, \ldots, n$
  - ri and Si are variables denoting the color components of f(x,y) and g(x,y) at any point (x,y).
  - n is the no of color components
  - {T1,T2,…..,Tn} is a set of transformation or color mapping functions.
- Note that n transformations combine to produce a single transformation T

Color Transformations

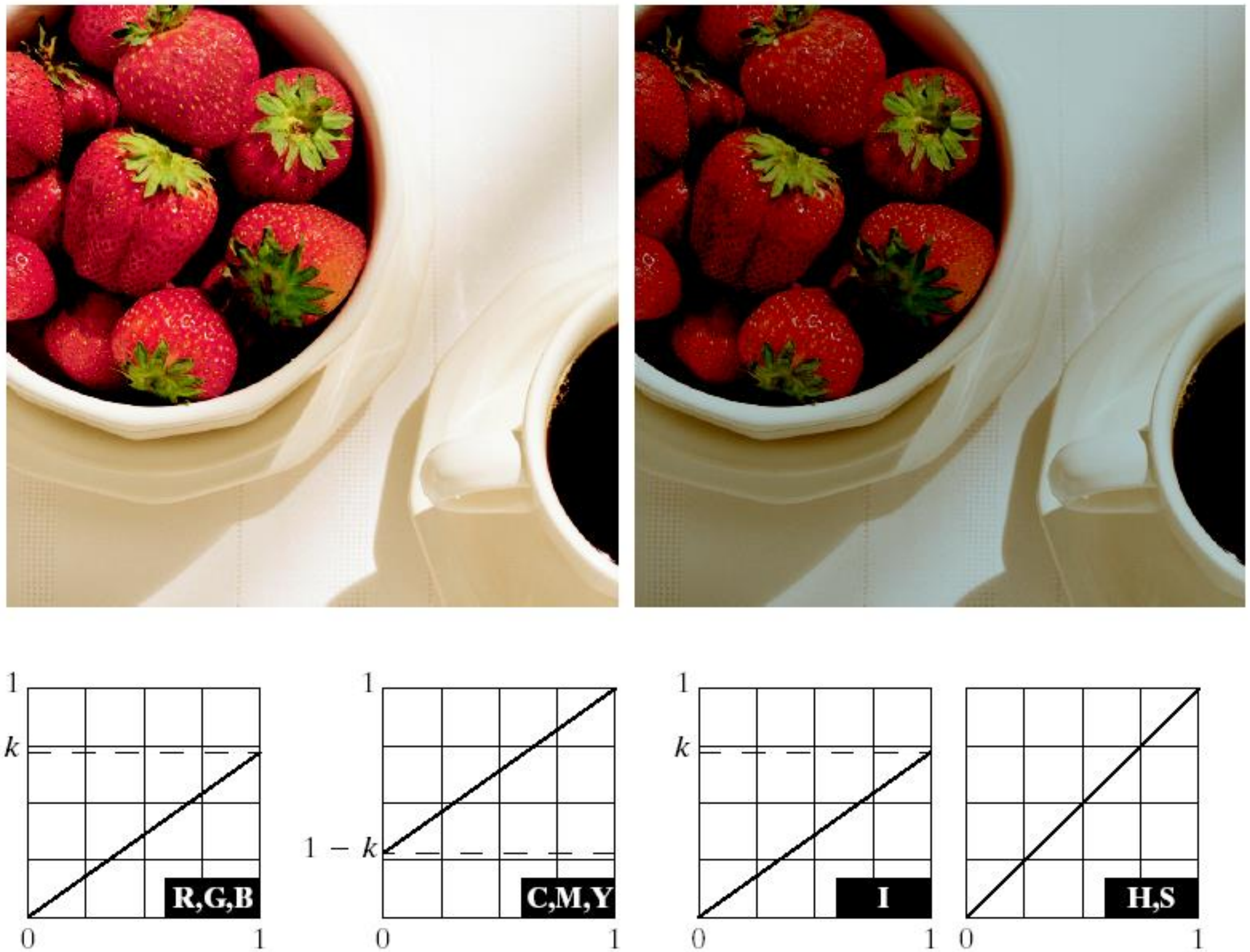
- The color space chosen determine the value of n.
- If RGB color space is selected then n=3 & r1,r2,r3 denotes the red, green and blue components of the image.
- If CMYK color space is selected then n=4 & r1,r2,r3,r4 denotes the cyan, Magenta, Yellow and black components of the image.
- Suppose we want to modify the intensity of the given image using $g(x,y)=k*f(x,y)$ where $0<k<1$

Full color

Color Transformations

- In HSI color space this can be done with the simple transformation
  - $s3=k*r3$
    where $s1=r1$ and $s2=r2$
  - Only intensity component r3 is modified.
- In RGB color space 3 components must be transformed:
  - $si=k*ri$ $i=1,2,3.$
- In CMY color space 3 components must be transformed:
  $si=k*ri + (1-k)$ $i=1,2,3.$
- Using k=0.7 the intensity of an image is decreased by 30%

Color Transformations



a b
c d e

FIGURE 6.31 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c)–(e) The required RGB, CMY, and HSI transformation functions. (Original image courtesy of MedData Interactive.)

Color Complement

- The hues opposite to one another on the Color Circle are called Complements.

- Color Complement transformation is equivalent to image negative in Grayscale images

- Color complements are useful for enhancing detail that is embedded in dark regions of a color image

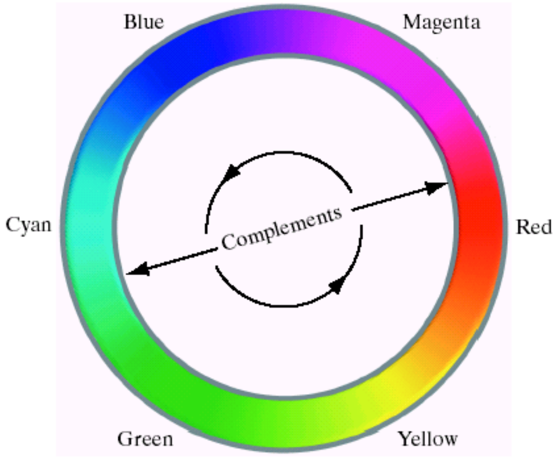
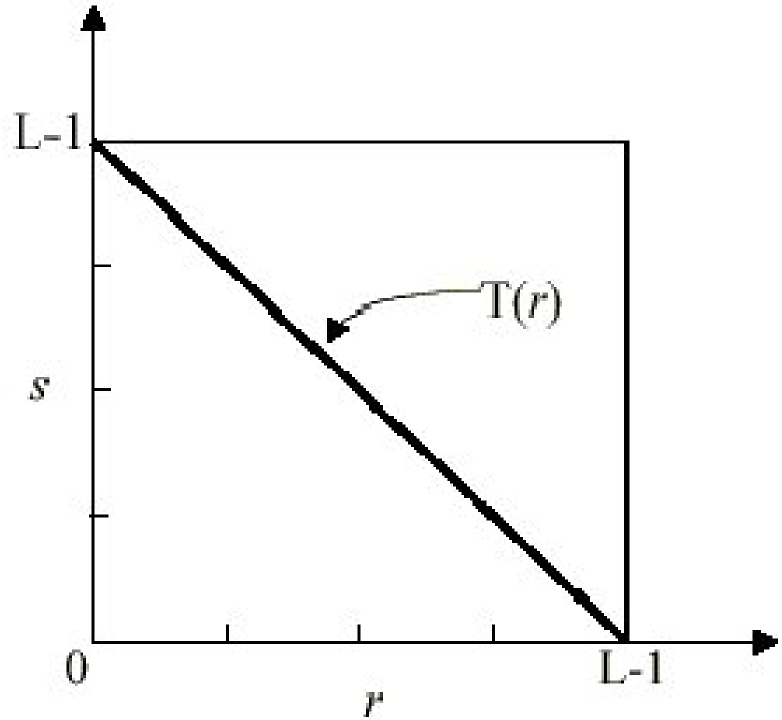


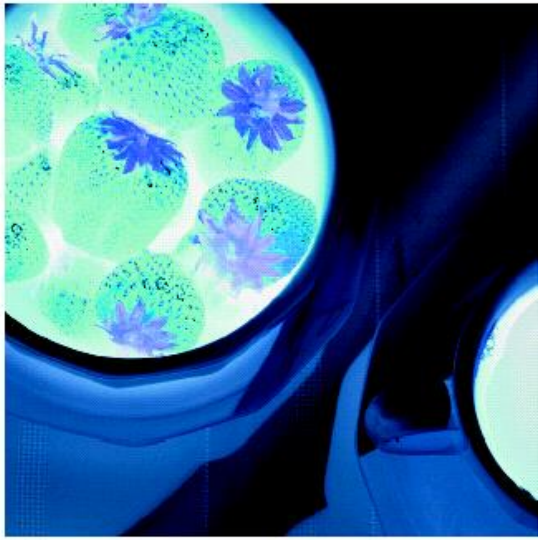
FIGURE 6.32 Complements on the color circle.

Color Complement

- **S=T(r)= L-1-r** **For Gray scale image**
- **Si=T(ri)=L-1-ri** **For Color image**

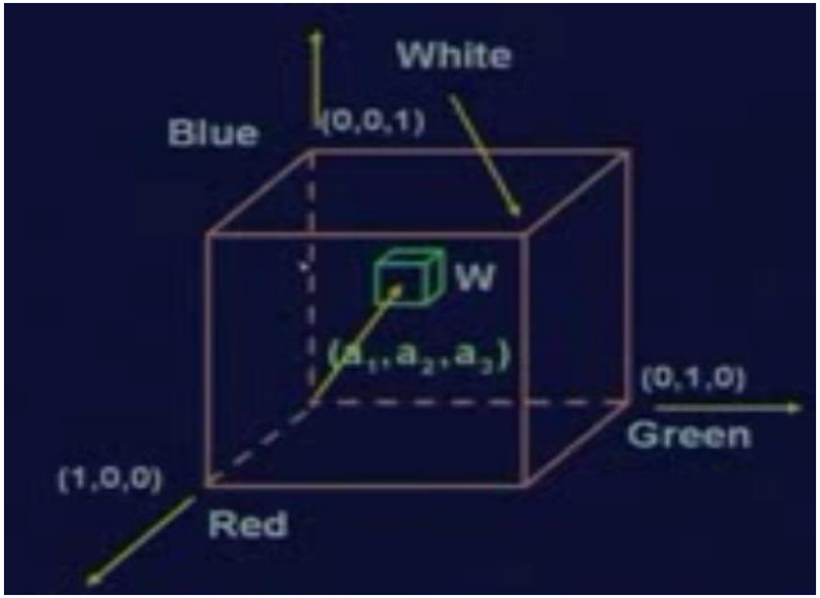
Where i=1,2,3





Color Slicing

- Highlighting a specific range of colors in an image is useful for separating objects from their surroundings.
- Display the colors of interest so that they are distinguished from background.
- One way to slice a color image is to map the color outside some range of interest to a non prominent neutral color (e.g., (R, G, B)=(0.5, 0.5, 0.5))



Color Slicing

- If the colors of interest are enclosed by a cube (or hypercube for n>3) of width W and centered at an average color (a) then the necessary set of transformation is:
  - **S_i = 0.5 if [| r_J - a_J | > W/2]** **for 1<=j<=3**
  - **S_i = r_i otherwise**

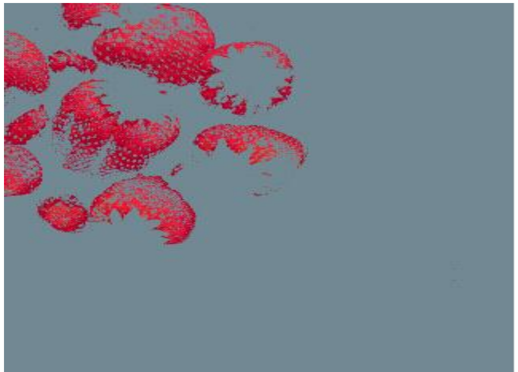
  Where i=1,2,3

- If a sphere is used to specify the colors of interest then

$$s_i = \begin{cases} 0.5 & if \quad \sum_{j=1}^{n} (r-a)^2 > R_0^2 \\ r_i & otherwise \end{cases}$$

- Forcing all other colors to the mid point of the reference color space.

Full color



Tone Correction

- Tonal Range: Low-key, Mid-key, High-key

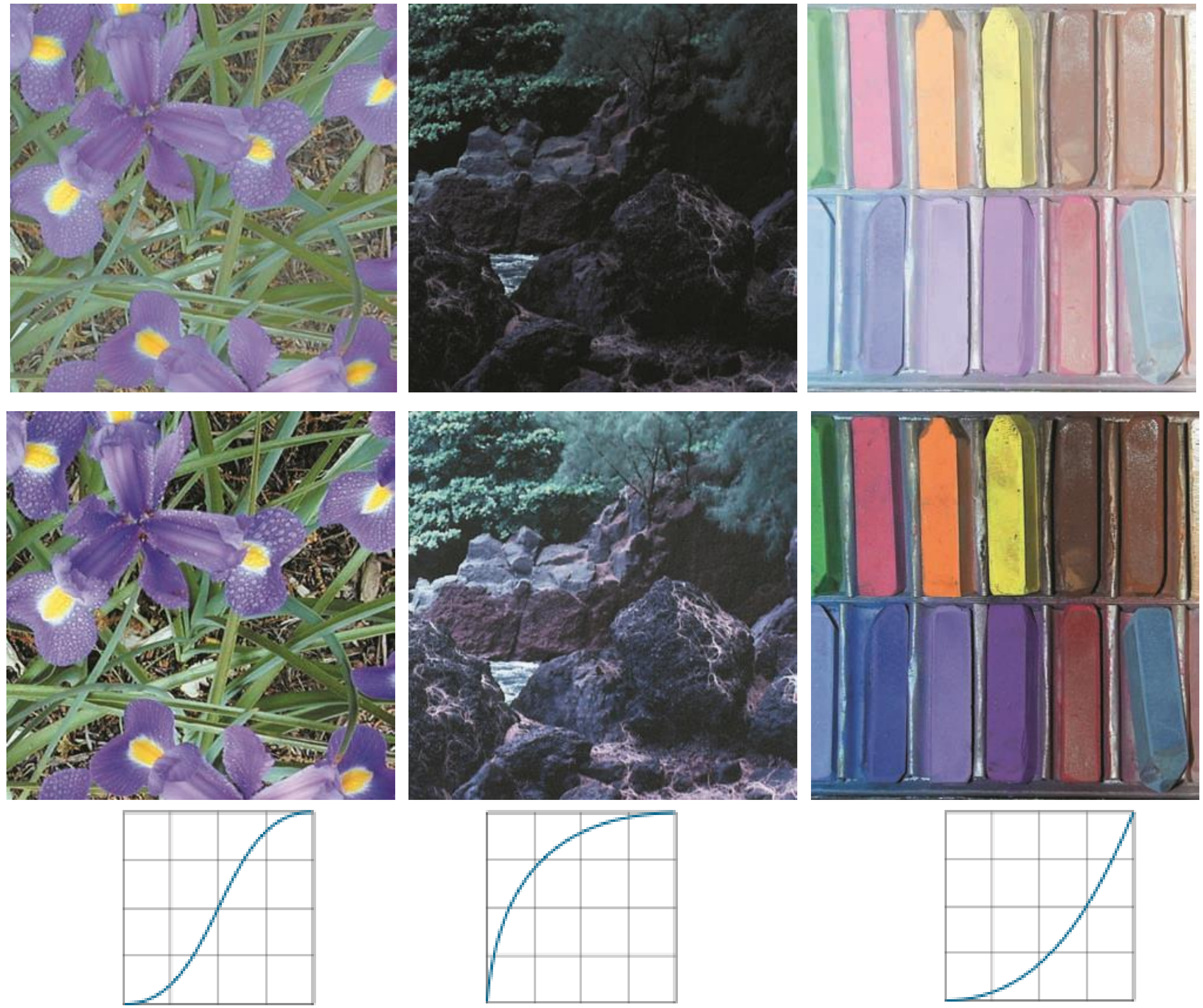


FIGURE 6.33 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

Color Correction

- When a color imbalance is noted, there are a variety of ways to correct it

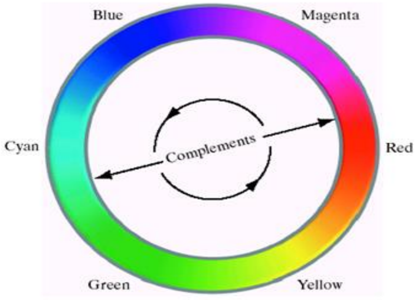
- When adjusting the color components of an image it is important to realize that every action affects the overall color balance of the image (Perception of one color is affected by its surrounding colors)

- Based on the color wheel, the proportion of any color can be increased by decreasing the amount of the opposite color in the image

- Similarly, it can increase by raising the proportion of two immediately adjacent colors or decreasing the percentage of the two colors adjacent to the complement

- Suppose for example there is an abundance of magenta in an RGB image, it can be decreased by

    - ✓ Reducing both red and blue or
    - ✓ Adding Green

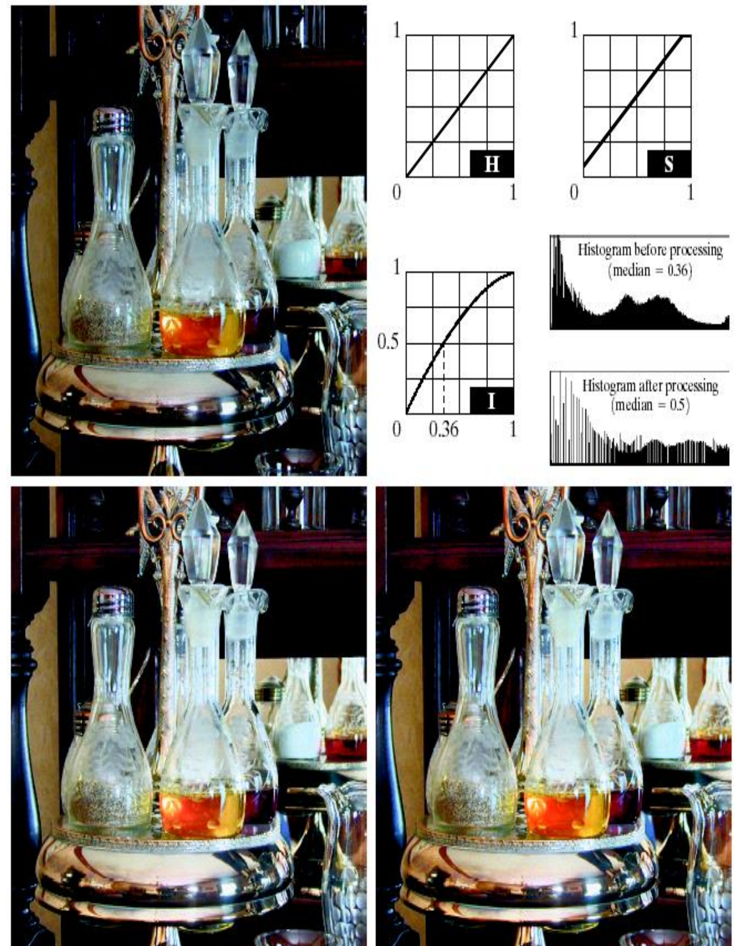


Histogram Processing

- Color images are composed of multiple components, however it is not suitable to process each plane independently in case of histogram equalization. This results in erroneous color.

- A more logical approach is to spread the color intensities uniformly, leaving the colors themselves(hue, saturation) unchanged.

- HSI approach is ideally suited to this type of approach.



a b
c d

FIGURE 6.37 Histogram equalization (followed by saturation adjustment) in the HSI color space.

Color Image Smoothing

- Color images can be smoothed in the same way as gray scale images, the difference is that instead of scalar gray level values we must deal with component vectors of the following form:

$$C(x,y) = \begin{pmatrix} C_R(x,y) \\ C_G(x,y) \\ C_B(x,y) \end{pmatrix} = \begin{pmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{pmatrix}$$

- The average of the RGB component vector in this neighborhood is:

$$\bar{C}(x,y) = \frac{1}{k} \sum_{(x,y) \in S_{xy}} C(x,y)$$

Color Image Smoothing

$$\bar{C}(x,y) = \begin{bmatrix} \frac{1}{k} \sum_{(x,y) \in S_{xy}} R(x,y) \\ \frac{1}{k} \sum_{(x,y) \in S_{xy}} G(x,y) \\ \frac{1}{k} \sum_{(x,y) \in S_{xy}} B(x,y) \end{bmatrix}$$

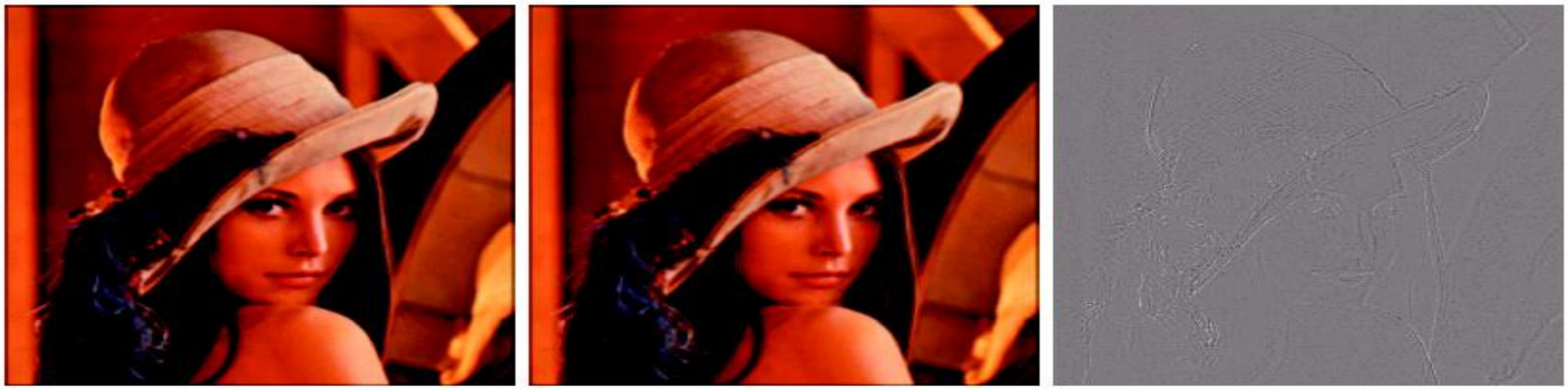
- We recognize the components of this vector as the scalar images that would be obtained by independently smoothing each plane of the starting RGB image using conventional gray scale neighborhood processing.
- Thus, we conclude that smoothing by neighborhood averaging can be carried out on a per color plane basis.

Color Image Smoothing

a b
c d

FIGURE 6.38 (a) RGB image. (b) Red component image. (c) Green component. (d) Blue component.

Color Image Smoothing



a b c

FIGURE 6.40 Image smoothing with a 5 × 5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

- The result of RGB and HSI are not identical as shown in the difference image of the RGB and HSI processed image
- This is due to the fact that average of the two pixels of differing color is a mixture of two colors, not either of the original colors (case of RGB)
- By Smoothing only the intensity component, the pixels maintain their original hue and saturation and thus their original colors

Color Image Sharpening

- The Laplacian of vector c is

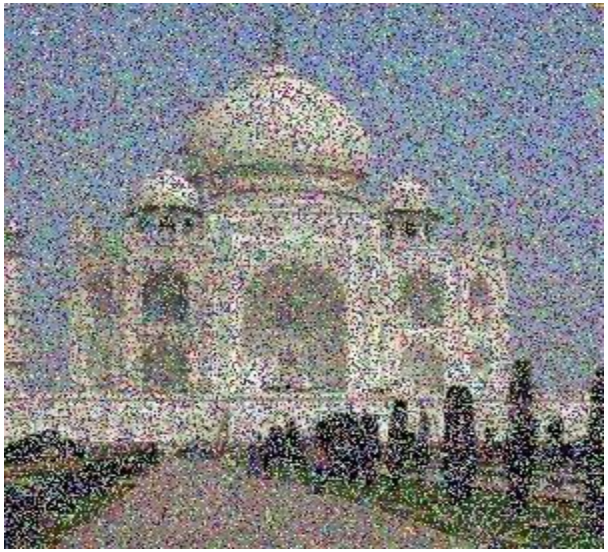
$$\nabla^2[c(x,y)] = \begin{bmatrix} \nabla^2 R(x,y) \\ \nabla^2 G(x,y) \\ \nabla^2 B(x,y) \end{bmatrix}$$

a b c

FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the intensity component and converting to RGB. (c) Difference between the two results.

Noise in Color Images

- We can employ image restoration approaches to remove the noise content of a color image if has the same characteristics in each color channel.
- It is possible for color channels to be affected differently by noise so in this case noise are removed from the image by independently processing each plane
- Remove noise by applying smoothing filters (e.g average, median) to each plane individually and then combine the result.



Color Image Compression

- Compression is the process of reducing or eliminating redundant and/or irrelevant information.
- A compressed image is not directly displayable it must be decompressed before input to a color monitor.
- In case if in a compressed image 1 bit of data represents 230 bits of data in the original image, then compressed image could be transmitted over internet in 1 minute as compared to original image which will take 4 hours to transmit.

Color Image Segmentation

- Color image segmentation is more effective in RGB color space.
- We make an estimate of average color we wish to classify pixels.
- Let average color be denoted by a in RGB space
- Objective is to assign this vector a to any region in color image based on specified distance measure (Euclidean distance)
- Let z be the arbitrary point in RGB space
- We say that z is similar to a if the distance between them is less than a specified threshold.

$$D(\mathbf{z}, \mathbf{a}) = \|\mathbf{z} - \mathbf{a}\|$$

$$= \left[(z_R - a_R)^2 + (z_G - a_G)^2 + (z_B - a_B)^2\right]^{\frac{1}{2}}$$

References

1. DIP by Gonzalez

For any query Feel Free to ask